touying-ufac-unofficial

Lecture slides for the Federal University of Acre (UFAC).

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https://github.com/lucaslrodri/touying-ufac-unofficial

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A Touying theme for lecture slides, following the visual identity of the Federal University of Acre. Slides are written with headings and a few Markdown-like shorthands for boxes, quotations, arrow lists and emphases. This manual is set with the rules of the theme itself.

Contents

1 Quick start 2
1.1 The information of the deck 2
1.2 The theme and the cover 3
1.3 Logos 5
1.4 Language 6
1.5 Dependencies 7
2 Slides and headings 8
2.1 Sections 8
2.2 Outline 9
2.3 Slides and continuations 9
2.4 Exercises and examples 10
2.5 Subtitles 13
2.6 Slide functions 14
3 Inline formatting and shorthands 20
3.1 Bold, italic and alert 20
3.2 Highlight and underline 20
3.3 Code chips 20
3.4 Colors by context 21
3.5 Colored emphases 21
3.6 Escapes 22
4 Lists and terms 23
4.1 Arrow lists 23
4.2 Bullet lists 23
4.3 Numbered lists 23
4.4 Terms 24
5 Boxes and quotes 25
5.1 Emphasis boxes 25
5.2 The box shorthand 26
5.3 Equation boxes 26
5.4 Quotations 27
6 Columns 29
7 Colors and logo 30
8 Tables, figures, math, code and icons 32
8.1 Tables 32
8.2 Figures and math 32
8.3 Code 32
8.4 Icons 34
9 Touying features 37
10 Index of definitions 39

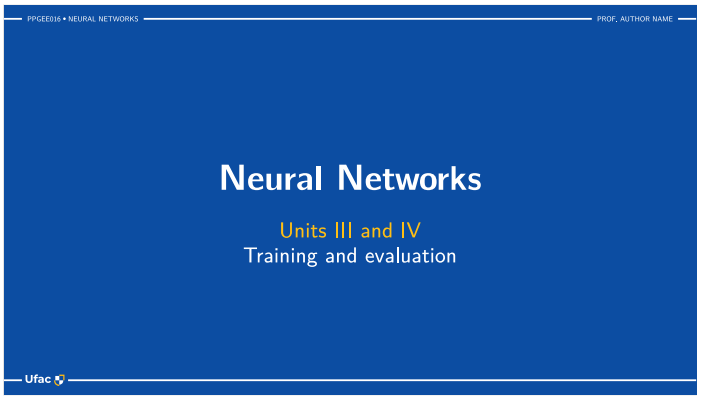
1 Quick start

A deck imports Touying and the theme, applies `ufac-theme` and usually starts with the cover:

```
#import "@preview/touying:0.7.4": *
#import "@preview/touying-ufac-unofficial:0.2.0": *
```

```
#show: ufac-theme.with(
  aspect-ratio: "16-9",
  lang: "en",
  config-info(
    subject: [Neural Networks],
    subtitle: [Units III and IV],
    title: [Training and evaluation],
    author: [Prof. Author Name],
    subject-code: [PPGEE016],
    counter-prefix: [1.],
  ),
)

#title-slide()
```



`typst init @preview/touying-ufac-unofficial:0.2.0` creates a deck from the package template, and `example/main.typ` shows the elements of the syntax in a short deck. From here on, every example shows the source on the left and the result on the right: a real slide when the feature belongs to the slide (headers, pills, continuations), otherwise the content alone, laid out by the same rules as in a deck.

1.1 The information of the deck

`config-info` is Touying's; the theme reads these fields:

Field	Where it appears
`subject`	first line of the cover (large, white) and left side of the footer
`subtitle`	second line of the cover, in yellow
`title`	third line of the cover, in white
`author`	top right of the cover and right side of the footer
`subject-code`	before the subject in the footer: "CODE · SUBJECT"
`counter-prefix`	prefix of the exercise and example numbers: `[1.]` gives "Exercise 1.N", `none` gives "Exercise N"

1.2 The theme and the cover

`ufac-theme()`

```
ufac-theme(
  aspect-ratio: "16-9" str
  lang: "pt-br" str
  exercise-name: auto auto content
  example-name: auto auto content
  exercise-icon: "pencil" none str content symbol
  example-icon: "beaker" none str content symbol
  part-name: auto auto content
  footer-left: auto auto content function
  footer-right: auto auto content function
  logos: auto auto array
  section-margin: false bool length
  ..args dictionary
  body content
) -> content
```

The theme. It sets the page (22pt New Computer Modern Sans, the UFAC header and footer) and every rule of the syntax, and takes Touying's configs as extra arguments.

```
#show: ufac-theme.with(
  aspect-ratio: "16-9",
  config-info(
    title: [Training and evaluation], subtitle: [Units III and IV],
    author: [Prof. Name], subject: [Neural Networks], subject-code: [PPGEE016],
    counter-prefix: [1.],
  ),
)
```

`config-info` carries `subject`, `subtitle`, `title`, `author`, `subject-code` and `counter-prefix`, the prefix of the exercise and example numbers ("Exercise 1.N"; `none` gives "Exercise N").

Argument

`aspect-ratio: "16-9"` — `str`

Any ratio accepted by Touying's `utils.page-args-from-aspect-ratio`: `"16-9"`, `"4-3"`, …

Argument

`lang: "pt-br"` — `str`

Language of the deck: `"pt-br"` (or `"pt"`), `"en"` or `"es"`, optionally with a region (`"en-US"`). It sets `text.lang`/`region` and the localized names below.

Argument

`exercise-name: auto` — `auto` | `content`

Name in the pill of exercise slides; `auto` is Exercício / Exercise / Ejercicio.

Argument

`example-name: auto` — `auto` | `content`

Name in the pill of example slides; `auto` is Exemplo / Example / Ejemplo.

Argument

`exercise-icon: "pencil"` — `none` | `str` | `content` | `symbol`

Icon before the name in the pill of exercise slides: the name of an `icon`, any content (an image or a symbol, centered on the capitals like the icons) or `none` for no icon. `exercise-slide(icon:)` changes it on one slide.

Argument

`example-icon: "beaker"` — `none` | `str` | `content` | `symbol`

Icon before the name in the pill of example slides, like `exercise-icon`.

Argument

`part-name: auto` — `auto` | `content`

Prefix of the section slide ("Part N:"); `auto` is Parte / Part / Parte.

Argument

`footer-left: auto` — `auto` | `content` | `function`

Left text of the footer; a function receives `self`. `auto` is the subject code, a dot and the subject (the dot only when both are given).

Argument

`footer-right: auto` — `auto` | `content` | `function`

Right text of the footer; a function receives `self`. `auto` is the author.

Argument

`logos: auto` — `auto` | `array`

Logos at the left of the footer and at the bottom of the cover, in this order, separated by a dot. Each one is content, a function that receives the colour of the footer (the primary colour on the slides, white on the cover), or `auto`, the UFAC logo (blue on the slides, white with the yellow shield on the cover). A logo should be about `0.64em` high, the height of the UFAC logo. `auto` is the UFAC logo alone; for the UFAC logo and a partner's, `(auto, color => image(bytes(read("lab.svg").replace("#000000", color.to-hex())), height: 0.64em))`.

Argument

`section-margin: false` — `bool` | `length`

Distance between the content of the section slides (what follows `=`, on the right) and the right edge of the page. `false` puts it flush with the edge, which suits a photo or a figure with a background; `true` keeps the page margin (`1.8em`, the same as the text), for an image without a background or an icon; a length sets the distance.

Argument

`..args` — `dictionary`

Touying configs, typically `config-info(..)`; also `config-common(..)`, `config-page(..)`.

Argument

`body` — `content`

The deck.

`title-slide()`

```
title-slide(
  ..body content
) -> content
```

Cover: blue page with the subject (large, white), the subtitle (yellow) and the title (white) of `config-info` centered, `footer-left`/`footer-right` in white at the top and the logos (`ufac-theme(logos:)`) in white at the bottom. The slide counter is frozen, so the cover does not count.

Content given to it, such as a partner's logo or the name of a program, goes below the titles, centered at the bottom of the cover, in white (`*bold*` and `#icon` too, as in a pill). It is placed, so the titles stay where they are without it; the space under them takes about a logo and two lines, and a taller content reaches them.

```
#title-slide[#image("lab-white.svg", height: 1.7em)]
```

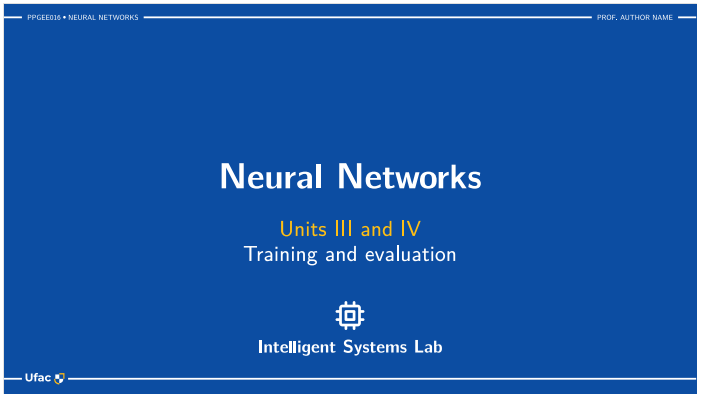
Argument

`..body` `content`

Content below the titles, `#title-slide[..]`; `#title-slide()` has none.

Content given to the cover goes below the titles, in white, as in a pill: a partner's logo, the name of a lab or of a program.

```
#title-slide[
  #icon("cpu", size: 1.6em) \
  *Intelligent Systems Lab*
]
```



1.3 Logos

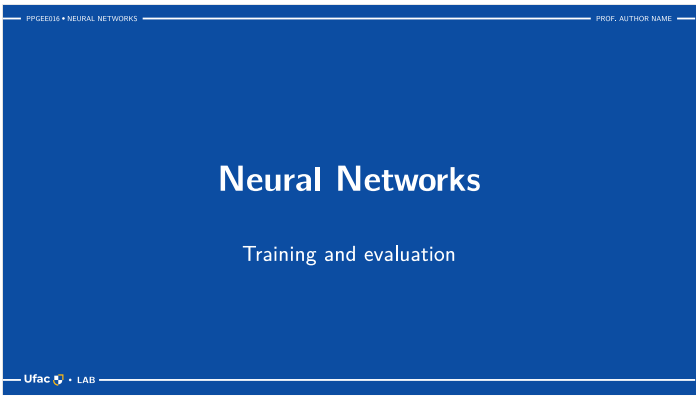
The footer and the cover show the UFAC logo. `logos` is the list of logos shown there, separated by dots: `(auto, x)` keeps the UFAC logo and adds `x` after it, `(x,)` replaces it. A logo is content, or a function that receives the colour of the footer (blue on the slides, white on the cover), so that one drawing serves both; `auto` stands for the UFAC logo. A logo about `0.64em` high matches the UFAC one. Outside the footer, as in a handout that goes with the slides, `ufac-logo-image` draws the UFAC logo in the same colours (Section 7).

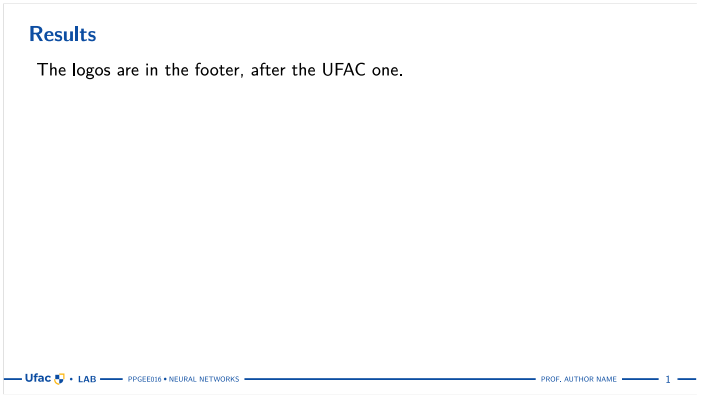
```
#show: ufac-theme.with(
  lang: "en",
  logos: (auto, color =>
  text(size: 0.5em, weight:
  "bold", fill: color)[LAB]),
  config-info(
    subject: [Neural
    Networks],
    title: [Training and
    evaluation],
    author: [Prof. Author
    Name],
    subject-code: [PPGEE016],
  ),
)

#title-slide()

== Results

The logos are in the footer,
after the UFAC one.
```





1.4 Language

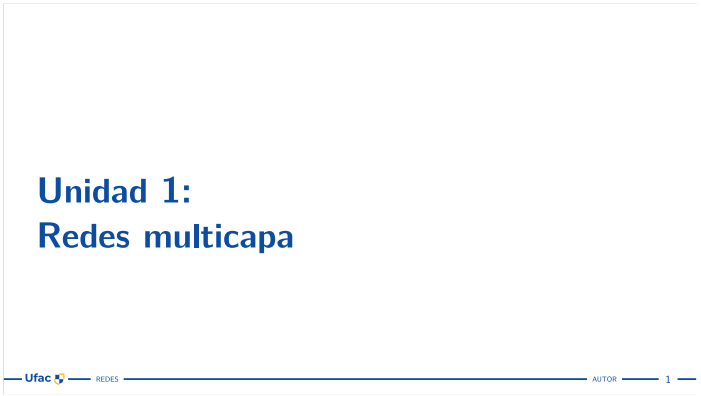
The theme speaks Portuguese by default (`lang: "pt-br"`): Exercício, Exemplo, Parte. `lang: "en"` or `"es"` switches the text language and those three names, and each name can still be overridden.

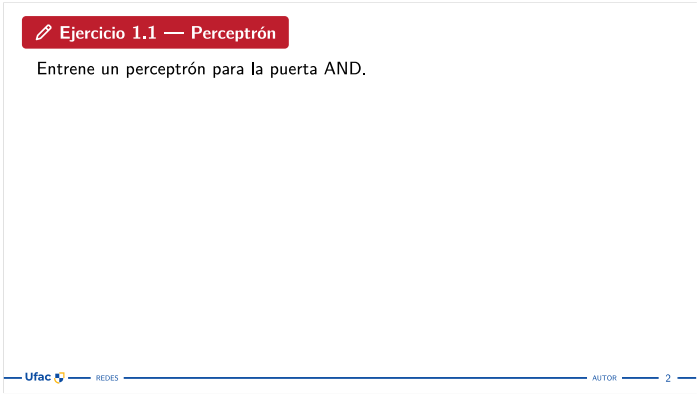
```
#show: ufac-theme.with(
  lang: "es",
  part-name: [Unidad],
  config-info(subject:
  [Redes], author: [Autor]),
)

= Redes multicapa

== Perceptrón <exercise>

Entrene un perceptrón para la
puerta AND.
```





1.5 Dependencies

The versions below are read from the sources of the package, so this table cannot go stale. They are fetched automatically by Typst. Newer major releases (touying 0.8, codly 2) require Typst 0.15; this version stays on the ones above, which also work with Typst 0.14.

Package	Version	Role
`codly`	`1.3.0`	code blocks; its functions are re-exported by the theme
`linguify`	`0.5.0`	localized names (Exercise, Example, Part)
`touying`	`0.7.4`	the slide engine; the deck imports it too, with the same version

Besides the packages, the theme needs **Typst 0.14.0 or later** and the font **New Computer Modern Sans** installed on the system (it is not bundled with Typst, unlike **New Computer Modern Math**, used for the equations). Building this manual also takes tidy 0.4.3, and the test suite runs on Tytanic 0.4.1.

2 Slides and headings

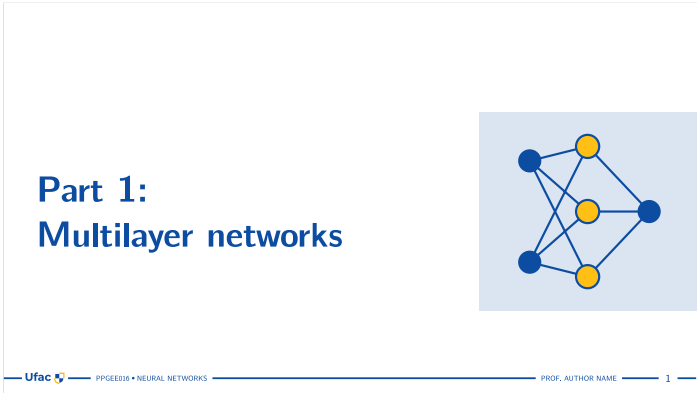
Headings drive the deck: no function call is needed for the ordinary slides.

Markup	Effect
`= Title`	section slide “Part N: Title”; what follows the heading goes to the right
`== Title`	new slide with a blue header, fitted to the width
`== Title <exercise>`	exercise slide: red pill with a pencil, “Exercise 1.N — Title”, numbered automatically
`== Title <example>`	example slide: green pill with a beaker, “Example 1.N — Title”, with its own counter
`=== Subtitle`	blue pill; no new slide; repeated at the top of every continuation page
`==== Subtitle`	the same pill, not repeated
`---`	alone on a line: next page of the same slide (same header, same number)

2.1 Sections

The content right after a `=` heading, typically an image, sits on the right, flush with the page edge, which suits a photo or a figure with a background.

```
= Multilayer networks
#image("figure.svg")
```



An image without a background or an icon looks better away from the edge: `section-margin: true` in `ufac-theme` keeps the page margin on every section slide, and a length sets another distance. On a single slide, `#pad(right: 1.8em)[#image(..)]` does the same.

```
#show: ufac-theme.with(
  lang: "en",
  section-margin: true,
  config-info(
    subject: [Networks],
    author: [Author Name],
  ),
)

= Multilayer networks
#image("network.svg")
```



2.2 Outline

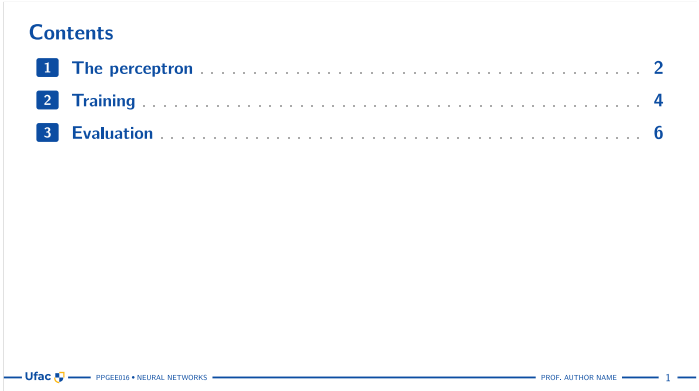
`#outline()` lists the parts of the deck, the `= Title` headings, with their numbers in small pills, the titles in bold blue and the number of the slide where each one starts, the number its footer shows; every line is a link to that slide. It usually takes a slide of its own, right after the cover. In a deck it stands for Typst's `outline`, which is still there as `std.outline`.

```
== Contents
#outline()

= The perceptron
== One neuron
A weighted sum and a threshold.

= Training
== Gradient descent
Small steps against the gradient.

= Evaluation
== Metrics
Accuracy, precision and recall.
```



outline()

outline(
) -> content

Outline of the deck: its parts (the `= Title` headings, the "Part N" of the section slides), one per line, with the number in a small pill, the title in bold blue and, after a dotted leader, the number of the section slide (the slide number of the footer, which skips the cover); every line is a link to that slide. In a deck it takes the place of Typst's `outline`, which is still `std.outline`. It goes on a slide of its own, usually the first after the cover:

```
== Contents
#outline()
```

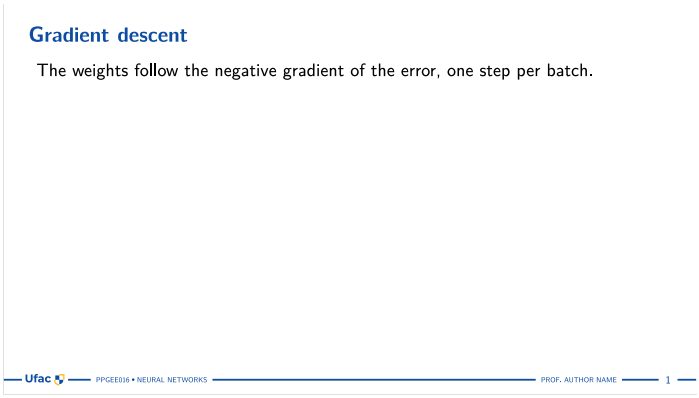
2.3 Slides and continuations

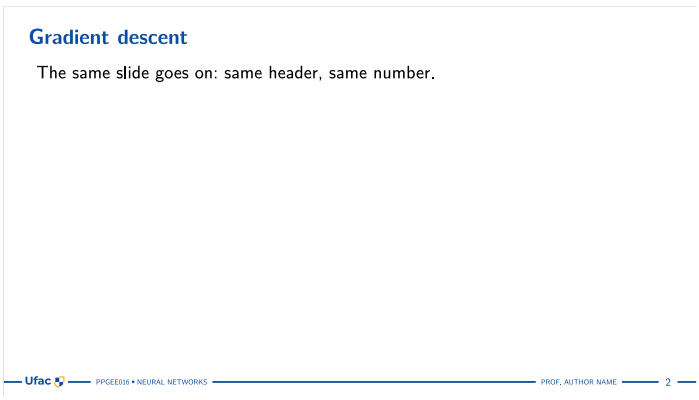
```
== Gradient descent

The weights follow the negative gradient of the error,
one step per batch.

---

The same slide goes on: same header, same number.
```





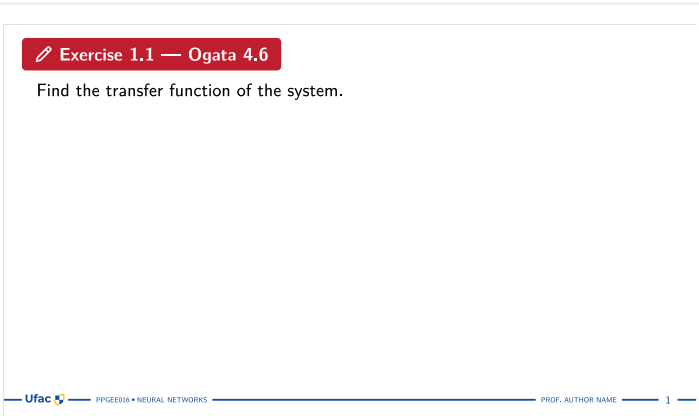
Caution

Touying breaks the page at *any* top-level em dash, so an em dash next to inline markup, as in `Text — *bold*`, also starts a new page. Use an en dash or wrap the paragraph in `#[ ]`.

2.4 Exercises and examples

A label on the `==` heading turns the header into a numbered pill: the icon of the kind (a pencil for exercises, a beaker for examples), the name, the number and, after a dash, the title. Any suffix gives the same slide and makes the label unique, so the slide can be cited: `@exercise-ogata` becomes "Exercise 1.N", bold, in the color of the kind and linked to the slide. `#exercise-slide` and `#example-slide` are the function forms, with `color` and `icon` options; they cannot be cited.

The icon is the name of an `icon`, any content (an image) or `none`, which leaves only the text. It is centered on the capitals of the text, whatever its height. `ufac-theme(exercise-icon:, example-icon:)` changes it in the whole deck; `icon:` changes it on one slide.



```
== Ogata 4.6 <exercise-ogata>

Find the transfer function of the system.

== XOR network <example>

Two hidden neurons are enough.

#exercise-slide(title: [Haykin 4.2], color: colors.primary, icon: "tools")[
  The function form, with another color and icon.
]

#example-slide(title: [Perceptron], icon: none)[
  Without icon: `icon: none`.
]

== Review

Solve @exercise-ogata before the next class.
```

Example 1.1 — XOR network

Two hidden neurons are enough.

Exercise 1.2 — Haykin 4.2

The function form, with another color and icon.

Example 1.2 — Perceptron

Without icon: `icon: none`.

Review

Solve Exercise 1.1 before the next class.

exercise-slide()

```
exercise-slide(
  title: [] content
  color: auto auto color
  icon: auto auto none str content
  body content
) -> content
```

Exercise slide: the header becomes a pill with a pencil, "Exercise 1.N — Title", numbered automatically. The function form of `== Title <exercise>`. Only the heading form can carry a unique label (`== Title <exercise-ogata>`) and be cited with `@exercise-ogata`, which shows "Exercise 1.N" linked to the slide.

```
#exercise-slide(title: [Ogata 4.6])[Find the transfer function.]
#exercise-slide(title: [Lab], color: colors.primary, icon: "tools")[Another color and icon.]
```

Argument

`title: []` — content

Text after the number, following a dash; empty omits the dash.

Argument

`color: auto` — auto | color

Color of the pill; `auto` is red (`tertiary`).

Argument

`icon: auto` — auto | none | str | content

Icon before the name: the name of an `icon`, any content (an image) or `none` for no icon; `auto` is the icon of the theme, `ufac-theme(exercise-icon:)` (a pencil).

Argument

`body` — content

Content of the slide; `---` inside it continues the same exercise.

example-slide()

```
example-slide(
  title: [] content
  color: auto auto color
  icon: auto auto none str content
  body content
) -> content
```

Example slide: a pill with a beaker, "Example 1.N — Title", with a counter independent from the exercises. The function form of `== Title <example>`; see `exercise-slide()`.

```
#example-slide(title: [XOR network])[Two hidden neurons are enough.]
```

Argument

`title: []` — content

Text after the number, following a dash; empty omits the dash.

Argument

`color: auto` — `auto` | `color`

Color of the pill; `auto` is green (`quaternary`).

Argument

`icon: auto` — `auto` | `none` | `str` | `content`

Icon before the name: the name of an `icon`, any content (an image) or `none` for no icon; `auto` is the icon of the theme, `ufac-theme(example-icon:)` (a beaker).

Argument

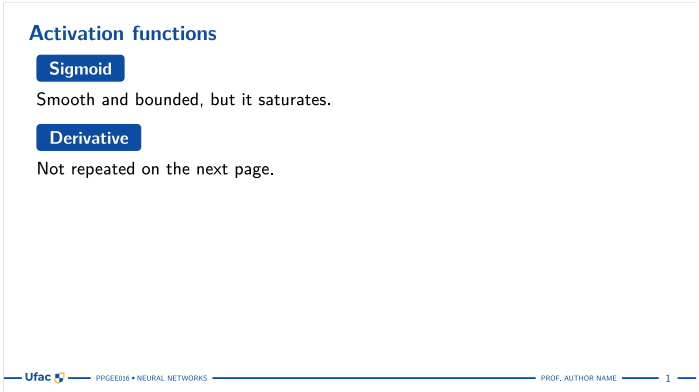
`body` — `content`

Content of the slide.

2.5 Subtitles

A `===` subtitle is repeated on the pages that continue the slide, unless the continuation starts with another `===`. That holds for `---`, for `#pagebreak()` and for content that overflows by itself.

```
1  == Activation functions
2
3  === Sigmoid
4
5  Smooth and bounded, but it
   saturates.
6
7  ==== Derivative
8
9  Not repeated on the next
   page.
10
11 ---
12
13 The `===` pill is repeated
   here.
```

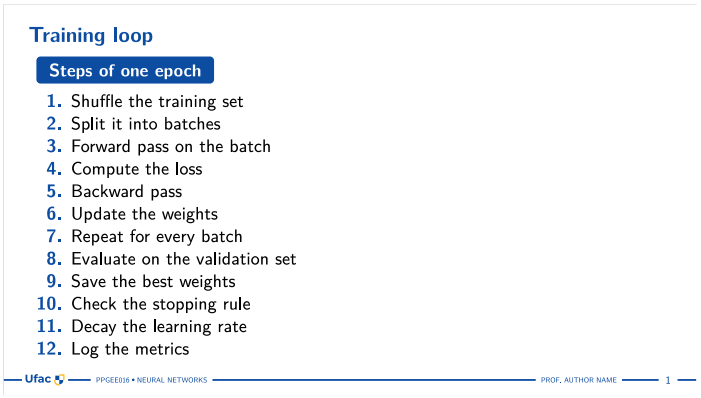


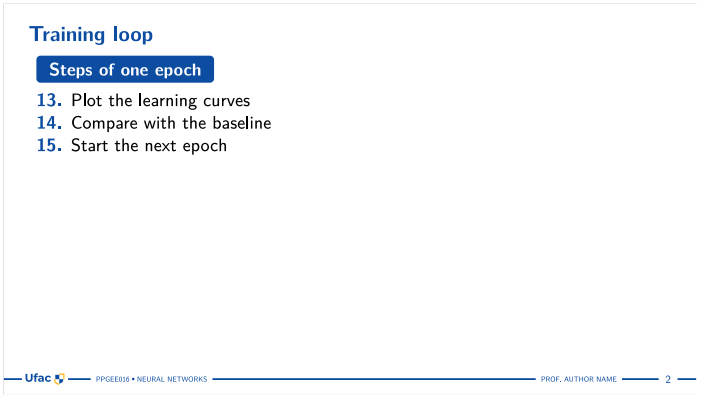


```
== Training loop

=== Steps of one epoch

+ Shuffle the training set
+ Split it into batches
+ Forward pass on the batch
+ Compute the loss
+ Backward pass
+ Update the weights
+ Repeat for every batch
+ Evaluate on the validation set
+ Save the best weights
+ Check the stopping rule
+ Decay the learning rate
+ Log the metrics
+ Plot the learning curves
+ Compare with the baseline
+ Start the next epoch
```





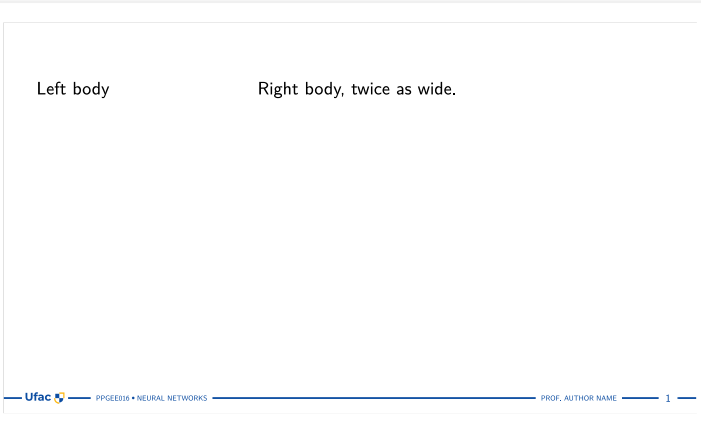
Only a top-level `===` is repeated; inside `#[ ]`, a column or a box the pill stays in the flow. `config-common(breakable: false)` makes Touying warn about overflow instead.

2.6 Slide functions

`#slide` is the slide that every `==` opens; calling it gives access to Touying's arguments, such as `composer`. `#empty-slide` drops the header and keeps the footer. `#title-slide()` is the cover, shown in Section 1; `#title-slide[..]` adds content below its titles.

```
#slide(composer: (1fr, 2fr))[
  Left body
][
  Right body, twice as wide.
]

#empty-slide[
  #align(center + horizon)[No header, footer kept.]
]
```

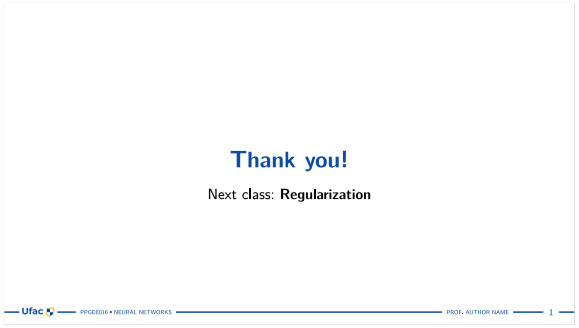




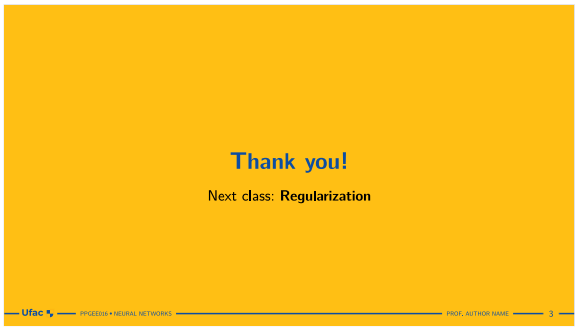
Closing and transition slides: `<empty-slide>`

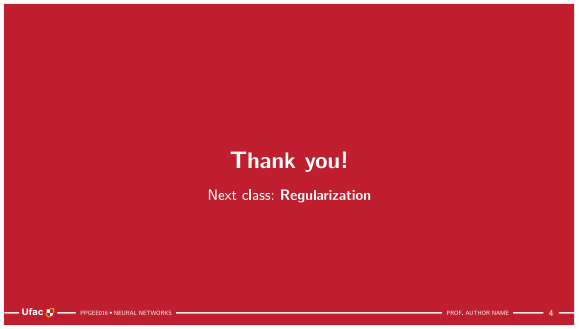
A `==` heading labelled `<empty-slide>` opens an `#empty-slide` with the heading as its title, large and centered, and the content centered below it. `<empty-slide-primary>` paints the page in that color of the palette; any key of `colors` works (`<empty-slide-secondary>`, `<empty-slide-tertiary-dark>`, …). On a dark color the text and the footer turn white, and bold, code and icons follow the pill convention; on a light one (`secondary`, the `-lighter` tones) they stay as on a white slide. The function form is `#empty-slide(color: colors.primary, title: [..])[..]`. The label styles the page its heading opens; a `---` continuation is a normal slide.

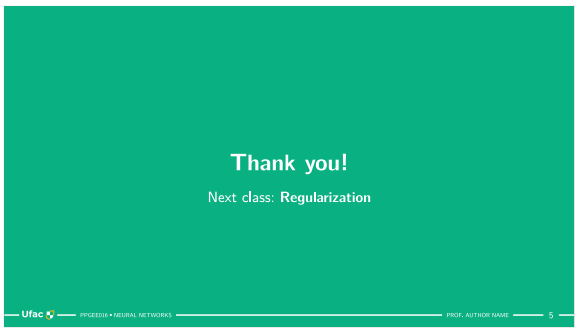
```
1  == Thank you! <empty-slide>
2  Next class: *Regularization*
3
4  == Thank you! <empty-slide-primary>
5  Next class: *Regularization*
6
7  == Thank you! <empty-slide-secondary>
8  Next class: *Regularization*
9
10 == Thank you! <empty-slide-tertiary>
11 Next class: *Regularization*
12
13 == Thank you! <empty-slide-quaternary>
14 Next class: *Regularization*
15
16 == Thank you! <empty-slide-neutral>
17 Next class: *Regularization*
```













slide()

```
slide(
  config: (:) dictionary
  repeat: auto auto int
  setting: body => body function
  composer: auto auto array function
  ..bodies content
) -> content
```

Default slide of the theme, with header and footer. Every `==` heading opens one, so it is rarely called by hand; the explicit form gives access to Touying's arguments.

A top-level `===` subtitle is repeated at the top of every continuation page of the slide, whether the page comes from `---`, from `#pagebreak()` or from content that overflows by itself, unless the continuation starts with another `===`. A subtitle nested in `#[ ]`, a column or a box is not repeated.

A `==` heading labelled `<empty-slide>` or `<empty-slide-primary>` (any color of the palette) opens an `empty-slide()` instead, with the heading as its title.

```
#slide(composer: (1fr, 2fr))[Left][Right]
```

Argument

`config: (:)` `dictionary`

Slide configuration: a `config-xxx` dictionary (merge several with `utils.merge-dicts`).

Argument

`repeat: auto` `auto` | `int`

Number of subslides; `auto` lets Touying count the animations.

Argument

`setting: body => body` `function`

Set/show rules applied to the slide body.

Argument

`composer: auto` `auto` | `array` | `function`

Layout of the bodies, e.g. `(1fr, 2fr)`, as in Touying's `side-by-side`.

Argument

`..bodies` `content`

The slide bodies.

empty-slide()

```
empty-slide(
  config: (:) dictionary
  repeat: auto auto int
  setting: body => body function
  composer: auto auto array function
  color: none none color
  title: none none content
  ..bodies content
) -> content
```

Slide without the header but with the footer (Touying's `empty-slide` removes both). Same arguments as `slide()`, plus the page color and an optional title. On a dark color (`primary`, `tertiary`, `quaternary`, `neutral`) the text and the footer turn white, and `*bold*`, code and icons follow the pill convention; on a light one (`secondary`, the `-lighter` tones) they stay as on a white slide.

The heading form, for a closing or transition slide: `== Title <empty-slide>` (white page) or `== Title <empty-slide-primary>`, `<empty-slide-secondary>`, … (any key of `colors`, including the tones, such as `<empty-slide-primary-dark>`). It is the same slide with the heading as the `title`: large and centered, with the content centered below it. The label styles the page its heading opens: a `---`

continuation is a normal slide, so a closing slide fits on one page (or each page is an `#empty-slide(color:, title:)`).

```
== Thank you! <empty-slide-primary>
Questions: `name@ufac.br`

#empty-slide(color: colors.tertiary)[#align(center + horizon)[*Break*: 15 minutes]]
```

Argument

`config: (:)` `dictionary`

Slide configuration, see `slide.config`.

Argument

`repeat: auto` `auto` | `int`

Number of subslides.

Argument

`setting: body => body` `function`

Set/show rules applied to the slide body.

Argument

`composer: auto` `auto` | `array` | `function`

Layout of the bodies.

Argument

`color: none` `none` | `color`

Color of the page, such as `colors.primary`; `none` keeps it white.

Argument

`title: none` `none` | `content`

Title, large and centered, with the bodies centered below it (the layout of the heading form); `none` leaves the bodies as they are.

Argument

`..bodies` `content`

The slide bodies.

new-section-slide()

```
new-section-slide(
  config: (:) dictionary
  level: 1 int
  numbered: true bool
  body content
) -> content
```

Section slide, created by every `= Title`: "Part N: Title" in blue on the left and the content right after the `=` (typically an `#image`) on the right, flush with the page edge. `ufac-theme(section-margin: true)` keeps the page margin on every section slide, for images without a background; `#pad(right: 1.8em)[#image(..)]` does the same on a single one. Not meant to be called by hand.

Argument

`config: (:)` `dictionary`

Slide configuration, see `slide.config`.

Argument

`level: 1` `int`

Heading level, passed by Touying.

Argument

`numbered: true` `bool`

Passed by Touying; the part is always numbered.

Argument

`body` `content`

Content that follows the `=` heading.

3 Inline formatting and shorthands

Most emphases have two forms: a function and a **shorthand**, a short piece of markup that the theme rewrites into the same function. The shorthands take plain text only (no markup inside) and none of them applies inside `raw`.

Function	Shorthand	Result
`#strong[text]`	`*text*`	black bold
`#emph[text]`	`_text_`	black italic
`#alert[text]`, `#primary[text]`	`~~text~~`	blue bold
`#highlight[text]`	`==text==`	light-yellow background
`#underline[text]`	`~text~`	strong yellow underline
`#secondary`, `#tertiary`, `#quaternary`		bold in yellow, red, green
`#emph-box[..]`	`> text`	box, Section 5
`$ #eq-box[$..$] $`	`> $ .. $`	equation box, Section 5
`#quote-box[..]`	`"text"`	quotation, Section 5
`#arrows[..]`	`-> item`	arrow list, Section 4

3.1 Bold, italic and alert

```
Black *bold* and _italic_.

Blue #alert[alert], the same as
~~this shorthand~~.
```

Black **bold** and *italic*.

Blue **alert**, the same as **this shorthand**.

3.2 Highlight and underline

The `==text==` shorthand cannot open a line (there it is a heading) and its text neither starts nor ends with a space; `#highlight` has no such limits and takes another fill, such as the `-lighter` tones of the palette.

```
A ==highlighted== word and a
#highlight(fill: colors.quaternary-
lighter)[green one].

A ~strong underline~ and
#underline[the function form]; `Fig.~1` keeps
its non-breaking space.
```

A highlighted word and a green one.

A strong underline and the function form `Fig.~1` keeps its non-breaking space.

3.3 Code chips

Inline `raw` is a gray chip. Inside a title or a box it follows the color of the container, and inside a pill it is inverted.

```
Call `fit(x, y)` first.

#emph-box(color: colors.tertiary)[
  Never call `predict` before `fit`.
]
```

Call `fit(x, y)` first.

Never call `predict` before `fit`.

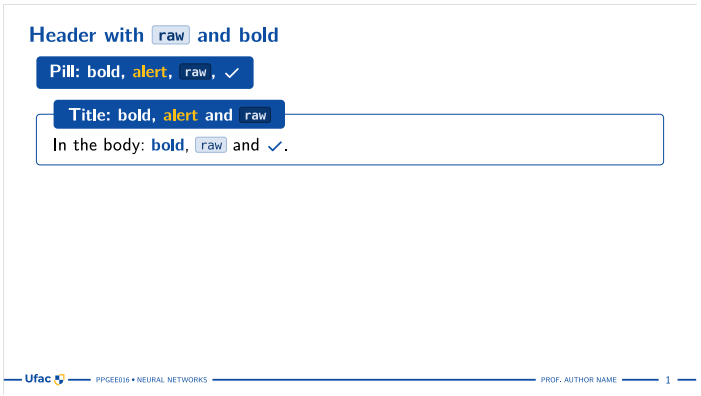
3.4 Colors by context

Bold text, alerts, highlights, underlines, arrows, chips and icons read the color of what contains them: the box color inside an `emph-box`, white or yellow inside a pill, where the pill color would hide them.

```
== Header with `raw` and *bold*

=== Pill: *bold*, ~~alert~~, `raw`, #icon("check")

#emph-box(title: [Title: *bold*, ~~alert~~ and `raw`])[
  In the body: *bold*, `raw` and #icon("check").
]
```



3.5 Colored emphases

Bold text named after the color families of Section 7. A typical use is a label that opens a sentence.

`primary()`

```
primary(
  body content
) -> content
```

Bold text in UFAC blue: the same as `#alert` (Touying's method) and as the `~~text~~` shorthand. Inside a pill (a `===` subtitle, an exercise header, the title of an `emph-box()`) it turns UFAC yellow, where blue would vanish.

```
A #primary[definition], the same as ~~this shorthand~~.
```

A **definition**, the same as **this shorthand**.

Argument

body — `content`

Text to emphasize.

`secondary()`

```
secondary(
  body content
) -> content
```

Bold text in UFAC yellow. It is weak on white: meant for short labels and dark backgrounds.

```
A #secondary[remark] in yellow.
```

A **remark** in yellow.

Argument

body — `content`

Text to emphasize.

tertiary()

```
tertiary(
  body content
) -> content
```

Bold text in red, for warnings and problems.

```
#tertiary[Caution:] the gradient vanishes.
```

Caution: the gradient vanishes.

Argument

body `content`

Text to emphasize.

quaternary()

```
quaternary(
  body content
) -> content
```

Bold text in green, for advantages and correct answers.

```
#quaternary[Correct:] the series converges.
```

Correct: the series converges.

Argument

body `content`

Text to emphasize.

3.6 Escapes

A backslash on the sign keeps a shorthand literal. An escaped `>` is only recognized when a word follows it.

```
\==not highlighted== and \~not underlined\~

\> not a box

-\> not a list

\"not a quotation\"
```

==not highlighted== and ~not underlined~

> not a box

-> not a list

"not a quotation"

4 Lists and terms

4.1 Arrow lists

Consecutive lines starting with `->` become one list with a blue arrow. The list has to start the paragraph: a blank line comes before the first `->`. An arrow in the middle of a line and the math arrow are left alone.

```
-> consequence
-> another consequence

Here a -> b stays text, and $a -> b$ is math.
```

→ consequence
→ another consequence

Here a -> b stays text, and $a \to b$ is math.

`arrows()`

```
arrows(
  color: colors.primary  color
  body  content
) -> content
```

Arrow lists with the arrow in another color, for everything inside the body. The `-> item` shorthand itself needs no function: consecutive lines starting with `->` become one list with a blue arrow (the box color inside an `emph-box()`). The list must start the paragraph, and `-\> item` escapes it.

```
#arrows(color: colors.quaternary)[
  -> green here
  -> and here
]
```

→ green here
→ and here

Argument

`color: colors.primary` `color`

Color of the arrows.

Argument

body `content`

Content with `-> item` paragraphs.

4.2 Bullet lists

Three levels of yellow markers: a filled square, a rotated hollow square and a small circle.

```
- first level
  - second level
    - third level
- first level again
```

- first level
  - second level
    - third level
- first level again

4.3 Numbered lists

Numbers are always blue and bold: `1.`, `a)`, `i.` by level. `#set enum(numbering: "a)")` starts at `a)` instead and keeps the colors. Typst only knows `+` and `1.` as enum markers in markup, so `a)` at the start of a line is text.

```
+ first
  + nested
    + deeper
+ second

#[
  #set enum(numbering: "a)")
  + starts at a)
    + then i.
]
```

1. first
   a) nested
      i. deeper
2. second

a) starts at a)
   i. then i.

4.4 Terms

```
/ Epoch: one pass over the training
set.
/ Batch: the samples of one update.
```

Epoch: one pass over the training set.
Batch: the samples of one update.

5 Boxes and quotes

5.1 Emphasis boxes

emph-box()

```
emph-box(
  color: colors.primary  color
  title: none  none  content
  icon: none  none  str  content
  body  content
) -> content
```

Emphasis box. Without a title it has a thin border; with a title, a badge sits over the top border. Inside the box the `-` markers, the `->` arrow, `*bold*`, `==highlight==` and `#underline` take the box color; an explicit `#arrows(color:)`, `#highlight(fill:)` or `#underline(stroke:)` wins. Colors by convention: primary for definitions, quaternary for advantages, tertiary for problems, secondary for remarks.

```
#emph-box(color: colors.quaternary,
title: [Pros])[
  - Fast and *simple*
  - Works with ==highlights==
]
#emph-box(color: colors.tertiary, icon:
"alert", title: [Caution])[
  An icon before the title.
]
```

Pros

- Fast and **simple**
- Works with highlights

⚠ Caution

An icon before the title.

Shorthand, always in the primary color: a paragraph whose lines start with `>`; a first line `> == Title` gives the title. Only inline content fits the shorthand. For a framed equation see `eq-box()`.

Argument

`color: colors.primary` — `color`

Color of the border and badge, and of the bold text, markers, arrows and highlights inside.

Argument

`title: none` — `none` | `content`

Badge over the top border; `none` gives the thin-border box (unless there is an `icon`).

Argument

`icon: none` — `none` | `str` | `content`

Icon at the start of the badge, before the title: the name of an `icon()`, any content (an image, a symbol) or `none`. It is centered on the capitals of the title, as in the exercise and example pills; without a title it is the whole badge. The badge has the same height with or without an icon.

Argument

`body` — `content`

Content of the box.

```
#emph-box[A definition, in the primary color.]

#emph-box(color: colors.tertiary, title: [Problem])[
  The gradient *vanishes* and ~stalls~.

  -> the arrow is red too
]

#emph-box(color: colors.quaternary, icon: "light-bulb", title: [Tip])[
  An `icon` before the title, as in the exercise pills.
]
```

A definition, in the primary color.

Problem

The gradient **vanishes** and ~~stalls~~.

→ the arrow is red too

Tip

An `icon` before the title, as in the exercise pills.

5.2 The box shorthand

A paragraph whose lines start with `>` is a box in the primary color; a first line `> == Title` gives the title.

```
> A box without title, with *bold* and $x^2$.

> == Definition
> A box with a title.
> The second line flows on.
```

A box without title, with **bold** and x^2.

Definition

A box with a title. The second line flows on.

Limits of the shorthand

Only inline content fits: lists and code blocks need `#emph-box`, and `>` before a block equation is an `eq-box`. A loose `>` at the start of a paragraph is read as a box: write `$a > b$`, or escape it (Section 3.6).

5.3 Equation boxes

eq-box()

```
eq-box(
  color: auto  auto  color
  body  content
) -> content
```

Equation box: a light frame around a key equation (the `key-eq` helper of the lecture decks). It goes **inside** the equation, like the `boxeq` of the bookly package, so the equation keeps its centering, spacing and numbering, and a part of an equation can be framed too. The body is written as math, between dollar signs; it is set in display style whatever the spacing around it. A box around the whole equation is padded generously, a box around a part of one tightly, and in both cases the frame holds tall fractions and the limits of sums.

It also works in a line of text or in an inline equation: there the body is in text style and the box does not change the line spacing.

```
The output #eq-box[$y = phi(v)$] and the
mean #eq-box[$1/N sum_i y_i$].
```

The output $\boxed{y = \varphi(v)}$ and the mean $\boxed{\frac{1}{N}\sum_i y_i}$.

```
$ #eq-box[$v = bold(w)^top bold(x)$] $

$ y = #eq-box(color: colors.quaternary)
[$phi(v)$] + b $
```

$$\boxed{v = \boldsymbol{w}^\top \boldsymbol{x}}$$
$$y = \boxed{\varphi(v)} + b$$

Shorthand, always in the primary color: `>` in front of a block equation, on the same line or alone on the line before it.

```
> $ E = 1/(2N) sum_i (y_i - hat(y)_i)^2
$
```

$$E = \frac{1}{2N} \sum_i (y_i - \hat{y}_i)^2$$

Argument

`color: auto` — `auto` | `color`

Base color of the fill and border; `auto` is blue, or the box color inside an `emph-box()`.

Argument

`body` — `content`

The equation, as math: `[$x = 2$]` or `($x = 2$)`.

Why the dollar signs inside

A content block inside an equation is markup, not math: `#eq-box[x = 2]` would be the upright text "x=2", and an underscore or an asterisk in it would not even parse. `eq-box` refuses such a body with a hint. `> $x$`, with an inline equation, is an ordinary box: the shorthand takes a block equation.

5.4 Quotations

`quote-box()`

```
quote-box(
  color: colors.secondary  color
  body  content
) -> content
```

Quotation box: a vertical bar on the left. Typst's `#quote(attribution: [Author])[…]` is shown as a `quote-box` too, with the attribution on the right.

```
#quote-box(color: colors.primary)[
  All models are wrong, but some are
  useful.
]
```

> All models are wrong, but some are useful.

Shorthand: a paragraph made only of `"quoted"` runs, one per line; each run becomes a box and the colors cycle secondary, primary, quaternary, tertiary. Quotes inside a sentence are left alone.

Argument

`color: colors.secondary` — `color`

Color of the bar.

Argument

`body` — `content`

The quotation.

```
"All models are wrong"
"but some are useful"

A sentence with a "quotation" inside.

#quote(attribution: [George Box])[
  All models are wrong.
]
```

> All models are wrong

> but some are useful

A sentence with a “quotation” inside.

> All models are wrong.
>
> — George Box

6 Columns

cols()

```
cols(
  columns: auto  auto  array
  gutter: 1em  length
  divider: false  bool
  ..bodies  content
) -> content
```

Columns: a thin wrapper over Touying's `cols` (a `grid`), with an optional divider. Touying's own `side-by-side` (no divider) is still available.

```
#cols(divider: true, columns: (1fr,
2fr))[
  Left
][
  -> the shorthands work
  -> inside a column
]
```

Left

→ the shorthands work
→ inside a column

Argument

`columns: auto` — `auto` | `array`

Column widths, as Touying accepts them: `auto` (equal widths) or an array such as `(1fr, 2fr)`.

Argument

`gutter: 1em` — `length`

Space between the columns.

Argument

`divider: false` — `bool`

A blue vertical bar (2.2pt) between the columns; the gutter is then used as inset.

Argument

`..bodies` — `content`

One content block per column.

```
#cols[Left][Middle][Right]
```

Left Middle Right

Touying's `side-by-side` (no divider) is still available, and `#slide(composer: (1fr, 2fr))[..][..]` lays out the bodies of a whole slide.

7 Colors and logo

Five families of seven tones. The light tones serve as fills of highlights and boxes, the dark ones as text on white.

	`-darkest`	`-darker`	`-dark`	base	`-light`	`-lighter`	`-lightest`
`primary`	#031329	#052349	#083671	#0C4DA2	#5582BE	#9EB8DA	#DBE4F1
`secondary`	#403005	#735609	#B2860E	#FFBF14	#FFD25A	#FFE5A1	#FFF5DC
`tertiary`	#30070B	#560D14	#851520	#BE1E2D	#D2626C	#E5A5AB	#F5DDE0
`quaternary`	#022C20	#044F3A	#067B5A	#09B081	#53C8A7	#9DDFCD	#DAF3EC
`neutral`	#000000	#2B2B2B	#555555	#808080	#AAAAAA	#D5D5D5	#FFFFFF

The tones are derived from the base in RGB: `darken(75%)`, `darken(55%)`, `darken(30%)`, the base, `lighten(30%)`, `lighten(60%)` and `lighten(85%)`. The neutral family follows Touying's default instead, from black to white.

```
#highlight(fill: colors.tertiary-
lighter)[red fill]
and #text(fill: colors.primary-dark)
[dark blue text].
```

red fill and dark blue text.

`colors` `dictionary`

The theme palette: the official UFAC colors plus a neutral gray, named after Touying's `config-colors` keys (`primary`, `secondary`, `tertiary`, `neutral`) plus `quaternary`. Each family has 7 tones, `<family>-darkest`, `-darker`, `-dark`, the base, `-light`, `-lighter` and `-lightest`, so `colors.primary-lighter`, `colors.quaternary-dark` etc. exist. The whole dictionary is passed to `config-colors`, hence the same keys are available as `self.colors.*`.

Family	Base	Role
`primary`	`#0C4DA2`	titles, `->` markers, tables, `#alert`/`#primary`, pills
`secondary`	`#FFBF14`	`-` markers, `#secondary`, quotes; `-lighter` is the highlight
`tertiary`	`#BE1E2D`	`#tertiary`, exercises
`quaternary`	`#09B081`	`#quaternary`, examples
`neutral`	`#808080`	`-darkest` is the bold/italic text, `-lightest` the background

The UFAC logo comes with the theme, as its SVG (`ufac-logo`) and as an image in the colours the theme uses (`ufac-logo-image`): blue lettering with the yellow shield in the footer, white lettering on the cover, and all white on a background where the yellow would not stand out, such as the cover of a handout that goes with the slides.

`ufac-logo` `str`

Official UFAC logo as an SVG string (yellow shield and blue lettering). `ufac-logo-image()` draws it, also in other colours (the footer, blue; the cover, white with the yellow shield).

The logo belongs to the Universidade Federal do Acre and is not covered by the license of the package.

```
#image(bytes(ufac-logo), height: 2em)
```



ufac-logo-image()

```
ufac-logo-image(
  height: 0.64em  length
  color: auto  auto  color
  shield: auto  auto  color
) -> content
```

The UFAC logo as an image, in the colours the theme uses: blue lettering with the yellow shield, as in the footer of the slides; the lettering in another colour, as on the cover (`color: white`, the shield still yellow); or all in one colour (`color: white, shield: white`), on a background where the yellow would not stand out. For a document that goes with the slides, such as a handout, or a logo next to it in `ufac-theme(logos:)`.

```
#ufac-logo-image(height: 2em)
#box(fill: colors.primary, inset: 0.4em,
  ufac-logo-image(height: 2em, color: white))
#box(fill: colors.primary, inset: 0.4em,
  ufac-logo-image(height: 2em, color: white, shield: white))
```



Argument

`height: 0.64em` `length`

Height of the logo; `0.64em` is its height in the footer.

Argument

`color: auto` `auto` | `color`

Colour of the lettering, the blue `#0c4da2` of the logo; `auto` keeps it.

Argument

`shield: auto` `auto` | `color`

Colour of the shield, the yellow `#fdb813` of the logo; `auto` keeps it.

8 Tables, figures, math, code and icons

8.1 Tables

The header row is blue with white text, and formulas in it turn white too. Code chips follow the background, as in the pills and boxes: in the header they are inverted (dark blue with light text). A cell filled with a light tone of a family of `colors` (`table.cell(fill: colors.secondary-lighter)`) is like a box of the family's `-dark` color, which colors its chips and `*bold*`; a cell filled with a dark tone (`colors.primary`, `colors.tertiary-dark`) is like a pill, with white text and inverted chips. White and colors outside the palette change nothing.

```
#table(
  columns: 3,
  table.header([Model], [Layers],
  [$eta$]),
  [MLP], [3], [0.1],
  [CNN], [8], [0.01],
)
```

Model	Layers	η
MLP	3	0.1
CNN	8	0.01

8.2 Figures and math

A figure caption shows its body only, without "Figure N". Math uses New Computer Modern Math, and block equations sit closer to the text than Typst's default. For a framed key equation, see `eq-box` in Section 5.

```
#figure(
  rect(width: 60%, height: 3em, fill:
  colors.primary-lightest),
  caption: [The caption, without a
  number.],
)

The update rule is
$ bold(w) <- bold(w) - eta nabla
E(bold(w)) $
```


The caption, without a number.

The update rule is

$$\boldsymbol{w} \leftarrow \boldsymbol{w} - \eta \nabla E(\boldsymbol{w})$$

8.3 Code

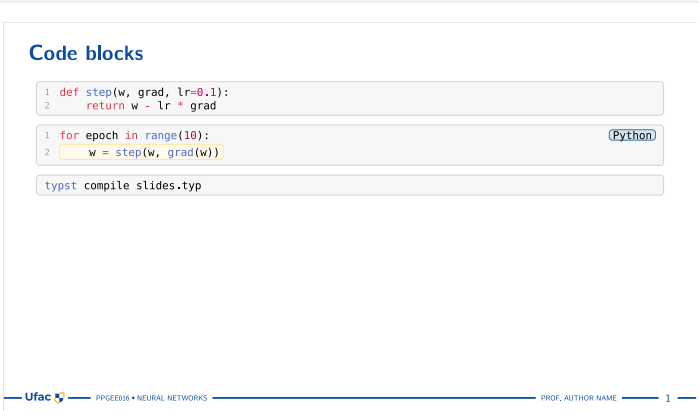
Inline `raw` is the chip of Section 3. Code blocks go through codly: a gray box, gray line numbers, no zebra, never split across pages unless `#local(breakable: true)[..]` (or `codly(breakable: true)`) allows it; the box then splits with the lines. The theme re-exports codly, and its `local` wraps codly's for per-block settings; `#codly-range(..)` and `#no-codly[..]` work as usual. The languages with a label are `python`, `typst`, `bash` and `text` (labelled "saída").

```
== Code blocks

```python
def step(w, grad, lr=0.1):
    return w - lr * grad
```

#local(display-name: true,
highlights: ((line: 2, fill:
colors.secondary-lighter),))[
  ```python
  for epoch in range(10):
      w = step(w, grad(w))
  ```
]

#local(number-format: none)[
  ```bash
  typst compile slides.typ
  ```
]
```



`local()`

```
local(
  body content
  ..args any
) -> content
```

Per-block code settings: codly's `local`, re-exported by the theme so that its code box and `raw` handling also apply inside (this `local` shadows codly's). Typical uses: `#local(number-format: none)[…]` hides the line numbers, `#local(display-name: true)[…]` shows the language label and `#local(highlights: ((line: 2, fill: colors.secondary-lighter),))[…]` highlights a line and `#local(breakable: true)[…]` lets a long block split across pages, box included. codly's `codly-range` and `no-codly` need no wrapper.

Argument

body `content`

Content with the code blocks.

Argument

..args `any`

Any argument of codly's `codly` function.

`code-languages` `dictionary`

Languages known to codly in code blocks. The `name` is what `#local(display-name: true)[…]` shows as a label; `text` is labelled "saída" (output), in Portuguese like the other user-facing defaults of the theme.

8.4 Icons

`icon()`

```
icon(
  name str
  color: auto auto color
  size: 1em length
  baseline: 25% relative
) -> content
```

An icon of the Octicons set (320 names), already wrapped in a `box` for inline use. An unknown name is a compile error.

```
#icon("check-circle", color: colors.quaternary) Done,
#icon("alert") attention and a big
#icon("rocket", size: 1.6em, color: colors.primary)
```

Done, attention and a big

Argument

`name` `str`

Name of the icon, e.g. `"check"`, `"alert"`, `"light-bulb"`, `"mark-github"`.

Argument

`color: auto` `auto` | `color`

Fill color. With `auto` it follows the context like bold text: white inside a pill (a `===` subtitle, an exercise header, the title of an `emph-box()`), the box or title color inside `emph-box()` and `==`, black elsewhere.

Argument

`size: 1em` `length`

Width and height of the icon.

Argument

`baseline: 25%` `relative`

Baseline of the box; the default aligns the icon with the text.

```
#emph-box(color: colors.tertiary)[
  #icon("alert") takes the box color.
]
```

takes the box color.

The 320 names, from Octicons (© GitHub Inc., MIT license):

accessibility, accessibility-inset, alert, alert-fill, apps
archive, arrow-both, arrow-down, arrow-down-left, arrow-down-right
arrow-left, arrow-right, arrow-switch, arrow-up, arrow-up-left
arrow-up-right, beaker, bell, bell-fill, bell-slash
blocked, bold, book, bookmark, bookmark-slash
briefcase, broadcast, browser, bug, cache
calendar, check, check-circle, check-circle-fill, checkbox
checklist, chevron-down, chevron-left, chevron-right, chevron-up
circle, circle-slash, clock, clock-fill, cloud


cloud-offline
code
code-of-conduct
code-review
code-square
codescan
codescan-checkmark
codespaces
columns
command-palette
comment
comment-discussion
container
copilot
copilot-error
copilot-warning
copy
cpu
credit-card
cross-reference
dash
database
dependabot
desktop-download
device-camera
device-camera-video
device-desktop
device-mobile
devices
diamond
diff
diff-added
diff-ignored
diff-modified
diff-removed
diff-renamed
discussion-closed
discussion-duplicate
discussion-outdated
dot
dot-fill
download
duplicate
ellipsis
eye
eye-closed
feed-discussion
feed-forked
feed-heart
feed-issue-closed
feed-issue-draft
feed-issue-open
feed-issue-reopen
feed-merged
feed-person
feed-plus
feed-public
feed-pull-request-closed
feed-pull-request-draft
feed-pull-request-open
feed-repo
feed-rocket
feed-star
feed-tag
feed-trophy
file
file-added
file-badge
file-binary
file-code
file-diff
file-directory
file-directory-fill
file-directory-open-fill
file-directory-symlink
file-moved
file-removed
file-submodule
file-symlink-file
file-zip
filter
fiscal-host
flame
fold
fold-down
fold-up
gear
gift
git-branch
git-commit
git-compare
git-merge
git-merge-queue
git-pull-request
git-pull-request-closed
git-pull-request-draft
globe
goal
grabber
graph
hash
heading
heart
heart-fill
history
home
horizontal-rule
hourglass
hubot
id-badge
image
inbox
infinity
info
issue-closed
issue-draft
issue-opened
issue-reopened
issue-tracked-by
issue-tracks
italic
iterations
kebab-horizontal
key
key-asterisk
law
light-bulb
link
link-external
list-ordered
list-unordered
location
lock
log
logo-gist
logo-github
mail
mark-github
markdown
megaphone
mention
meter
milestone
mirror
moon
mortar-board
move-to-bottom
move-to-end
move-to-start
move-to-top
multi-select
mute
no-entry
north-star
note
number
organization
package
package-dependencies
package-dependents
paintbrush
paper-airplane
paperclip
passkey-fill
paste
pencil
people
person
person-add
person-fill
pin
pin-slash
pivot-column
play
plug
plus
plus-circle
project
project-roadmap
project-symlink
project-template
pulse
question
quote
read
redo
rel-file-path
reply
repo
repo-clone
repo-deleted
repo-forked
repo-locked
repo-pull
repo-push
repo-template
report
rocket
rows
rss
ruby
screen-full
screen-normal
search
server
share
share-android
shield
shield-check
shield-lock
shield-slash
shield-x
sidebar-collapse
sidebar-expand
sign-in
sign-out
single-select
skip
skip-fill
sliders
smiley
sort-asc
sort-desc
sparkle-fill
sponsor-tiers
square
square-fill
squirrel
stack
star
star-fill
stop
stopwatch
strikethrough
sun

sync	tab-external	table	tag	tasklist
telescope	telescope-fill	terminal	three-bars	thumbsdown
thumbsup	tools	tracked-by-closed-completed	tracked-by-closed-not-planned	trash
triangle-down	triangle-left	triangle-right	triangle-up	trophy
typography	undo	unfold	unlink	unlock
unmute	unread	unverified	upload	verified
versions	video	webhook	workflow	x
x-circle	x-circle-fill	zap	zoom-in	zoom-out

9 Touying features

The theme sits on Touying and leaves its features alone. Animations work on every element of the theme: covered content stays faintly visible under a veil in the page color.

```
== Animations

First this.

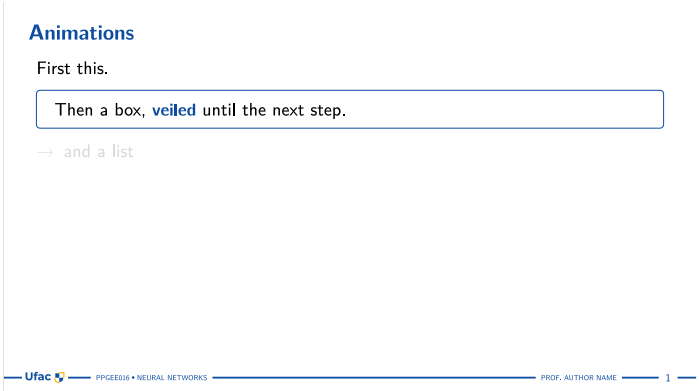
#pause

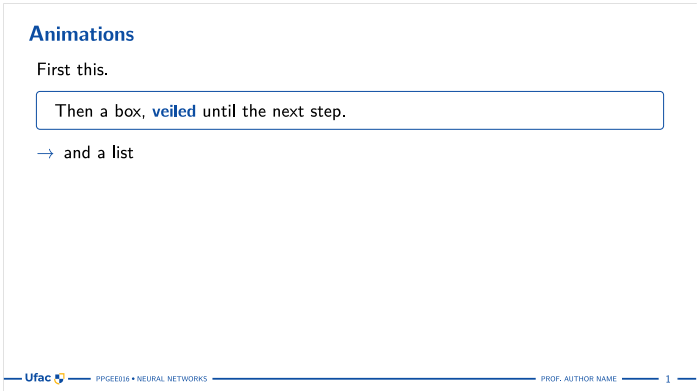
> Then a box, *veiled* until the next step.

#pause

-> and a list
```







Keep `#pause` and `#uncover`/`#only` in separate paragraphs: within one paragraph Touying reorders the covered part on the first subslide.

Available without any wrapper from the theme:

Feature	Functions
Animations	`#pause`, `#meanwhile`, `#jump`, `#uncover`, `#only`, `#alternatives`, `#item-by-item`, `#effect`, `#touying-raw`

Feature	Functions
Notes and handouts	`#speaker-note` (it attaches to the previous slide), `#handout-only`, `config-common(handout: true)`
Layout	`#slide(composer: (1fr, 2fr))`, `#cols`, `components.adaptive-columns`, `components.full-width-block`, `utils.fit-to-width`, `utils.fit-to-height`
Overflow	`config-common(breakable: false, clip:, detect-overflow:)`
Reuse	`#touying-recall`, `touying-get-config`, `components.progressive-outline`

Slides of other Touying themes that this one does not have: `outline-slide` (here, a slide with `#outline()`, see Section 2), `focus-slide`, `ending-slide` and `matrix-slide`.

10 Index of definitions

The reference of every function and variable sits in the chapter that introduces it; it is generated from the doc-comments of the package with tidy. Names that start with an underscore are internal and left out.

`ufac.typ`

`empty-slide()` p. 17
`example-slide()` p. 12
`exercise-slide()` p. 12
`new-section-slide()` p. 18
`slide()` p. 16
`title-slide()` p. 5
`ufac-theme()` p. 3

`components.typ`

`arrows()` p. 23
`cols()` p. 29
`emph-box()` p. 25
`eq-box()` p. 26
`icon()` p. 34
`local()` p. 33
`outline()` p. 9
`primary()` p. 21
`quaternary()` p. 22
`quote-box()` p. 27
`secondary()` p. 21
`tertiary()` p. 22

`constants.typ`

`code-languages` p. 33
`colors` p. 30
`ufac-logo` p. 30
`ufac-logo-image()` p. 31